

# Wired Portable Smart Charger with Hybrid Power Source and Intelligent Load Management System

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**Abstract**— The increased reliance on portable electronic devices in our daily lives has seen the efficient and multi-device charging become a practical need and not a luxury. This piece of work explains the design and development of a Wired Portable Smart Charger with Hybrid Power Source and Intelligent Load Management System - a small charging station designed to make this trouble a little easier in a significant and sustainable manner. The suggested system will combine several devices and powering requirements in a single station with three USB-C output ports. The former has an input of 28V and can provide up to 100W to power devices that are power-hungry, such as laptops. The second port is 22V, and has a maximum output of 65W and is aimed at tablets and mid-range electronics. The third port offers almost 18V up to 30W, powered by the main wired supply and a built-in solar panel, and thus is appropriate to low-power wearables and personal care devices. Such a hybrid power structure renders the system as well suited to turn up in a city-house or off-the-grid settings. The distinguishing feature of this system is its smart load control, real-time monitoring of voltage, current, power and energy source selection with CO<sub>2</sub> emission indication.

**Keywords**— Portable Smart Charger, Hybrid Power Source, Intelligent Load Management System, USB-C output ports, solar panel, CO<sub>2</sub> emission indication

## I. INTRODUCTION

The consumer electronics landscape today encompasses a remarkably diverse range of devices. Smartphones and tablets are obvious staples, but laptops, smartwatches, wireless earbuds, electric toothbrushes, and electric shavers are now equally embedded into daily routines. Each of these devices demands a different charging voltage and power level. Laptops may need 100W or more, tablets typically require around 65W, and smaller wearables and personal care gadgets comfortably operate at 30W or below. A charging solution that can intelligently serve all of these needs from a single station without compromising speed, safety, or stability would represent a meaningful and practical advancement over what is currently available.

Existing multi-port chargers do attempt to address this need, but they come with notable limitations. Most distribute a fixed total wattage across all ports regardless of what is

actually connected, which often means that devices charge slowly when multiple ports are in use. Very few incorporate any real intelligence about load conditions, and almost none integrate renewable energy sources. For users in remote areas, frequent travelers, or anyone conscious about their carbon footprint, these limitations matter.

Renewable energy and solar power in particular has matured considerably at the small-scale level. Compact photovoltaic panels and affordable power management ICs have made it genuinely feasible to embed solar harvesting into a portable device. When combined with a primary wired power source, a hybrid architecture emerges that is more resilient, more sustainable, and more versatile than either source alone. This kind of hybrid design is especially valuable in outdoor and off-grid scenarios, where grid access may be limited or entirely unavailable.

Equally important and often overlooked in engineering design is the human side of energy awareness. Most people have little visibility into how much energy their devices are consuming, or what the environmental cost of that consumption looks like. A system that not only manages energy efficiently but also communicates that information clearly through real-time displays of voltage, current, power flow, energy source selection, and even an estimated CO<sub>2</sub> emission indicator transforms a charger from a passive utility into an interactive and environmentally conscious tool. This kind of feedback has the potential to shift user behavior over time, nudging people toward more sustainable energy habits simply by making the invisible visible.

With all of these considerations in mind, this paper presents the design, development, and evaluation of a Wired Portable Smart Charger with Hybrid Power Source and Intelligent Load Management System. The proposed station integrates three USB-C ports with distinct power profiles 28V/100W, 22V/65W, and a hybrid solar-assisted 18V/30W port governed by an intelligent load management module that monitors and distributes power dynamically in real time. The system also incorporates a live monitoring interface displaying voltage, current, power consumption per port, active energy source, and a CO<sub>2</sub> savings indicator to foster environmental impact awareness which distinguishes from the rest.

## II. LITERATURE REVIEW

Here, [1] proposed an effective blockchain-based intelligent EV charging station management system (CSMS) that integrates a IoT technology with a double-layered centralized and decentralized optimization architecture, where the upper layer performs optimal charging and discharging scheduling at 15-minute intervals while the bottom layer executes distributed control at each charging pile every 5 minutes. In [2] they presented a solar-powered mobile phone charger designed to address the persistent challenge of device charging in areas with limited or no access to conventional power sources, using two 6V/80mA solar panels along with an LM317 voltage regulator and a Zener diode-based overcharge protection circuit to deliver a stable 5V output. Similarly, [3] presented a comprehensive review of Home Energy Management Systems (HEMS), highlighting how these systems use demand response tools to optimize household energy consumption by creating intelligent schedules based on energy costs, load profiles, and consumer comfort. [5] conveys us that Home Energy Management Systems (HEMS) have emerged as a key solution to optimize residential energy consumption through intelligent scheduling and control of household appliances. Here, it also showcases that recent studies focus on integrating renewable energy sources, smart grids, and advanced communication technologies to improve efficiency and reduce costs. [4] demonstrates how hardware components (CPU, GPU, display, sensors), signal modules (Wi-Fi, Bluetooth, GPS, cellular networks), software applications, and user behavior patterns collectively contribute to the utmost percentage of energy drain in smartphones. In [8] it highlights that battery management systems (BMS) are essential for ensuring safe and efficient operation of modern energy storage systems across applications ranging from portable devices to electric vehicles. Studies emphasize key BMS functions such as state estimation, cell balancing, and protection against overcharge, overdischarge, and thermal issues. Integration of renewable energy sources and storage systems further enhances energy efficiency, reduces dependency on the grid, and improves system reliability which is highly efficient. These works collectively highlight the importance of intelligent scheduling and optimization in achieving cost-effective and sustainable smart home energy management which was proposed in [7]. In [10], recent research on electric vehicle (EV) charging highlights the limitations of conventional fixed charging stations, such as long waiting times which consumes a lot of time, high infrastructure costs which leads to high costs, and spatial constraints. Mobile charging systems have emerged as a flexible and an alternative option, enabling on-demand energy delivery and reducing range anxiety. Additionally, [9] demonstrates that the DC-DC converters and multi-stage battery charging strategies (bulk, absorption, and float) are here employed to ensure efficient and safe battery charging. However, here [6] tells us that recent research highlights the integration of renewable energy sources, particularly photovoltaic systems, with electric vehicle (EV) charging infrastructure to support sustainable mobility. Here, it shows that both wired and wireless charging technologies have been extensively analyzed and have been extensively used, with wireless power transfer (WPT) emerging as a promising and a trustworthy solution despite challenges such as efficiency loss, alignment issues, and system complexity which constitutes as a loss but gives us knowledge on this aspect.

## III. METHODOLOGY

### A. Overall System Overview

The system receives power primarily from an AC source, which is distributed across three independent USB-C output channels. In addition, a solar panel is integrated into the system to support hybrid energy operation. The third port is uniquely designed to draw power from both AC and solar inputs, enabling simultaneous dual-source utilization.

Each output port is configured for a specific power rating:

1. Port 1: 28V, up to 100W (high-power devices such as laptops)
2. Port 2: 22V, up to 65W (medium-power devices such as tablets)
3. Port 3: 18V, up to 30W (low-power devices with hybrid solar support)

To enable real-time monitoring, each port is equipped

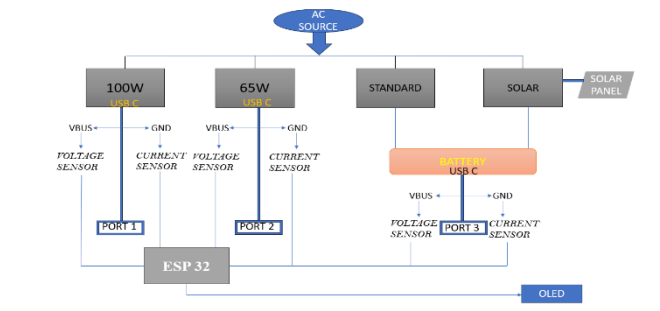


Fig.1. System architecture of wired portable smart charger with hybrid power source and intelligent load management system

with-voltage sensors and current sensors.

These sensors continuously measure electrical parameters, which are fed into the ESP32 microcontroller. The collected data is used to compute instantaneous power consumption and monitor system performance.

A decision-making mechanism is implemented to optimize energy usage between the AC source and the solar panel. The controller:

- Monitors solar availability and load conditions.
- Prioritizes solar energy when sufficient power is available.
- Switches or supplements with AC power when solar input is insufficient.

This hybrid approach enhances energy efficiency and promotes renewable energy utilization.

The system incorporates a CO<sub>2</sub> emission indicator, which estimates the environmental impact of energy consumption based on the proportion of solar versus grid energy used. This information, along with voltage, current, and power data, is displayed on an OLED interface, providing real-time feedback to the user.

All modules like power distribution, sensing units, control logic, and display are integrated into a compact and portable system which also works as wired and wireless due to the renewable form of energy like solar energy is used.

### B. 100W, 65W and standard charger design

The proposed system is implemented as a compact multi-output power supply that integrates three independent charger modules—rated at 100W, 65W, and a standard low-power output—on a single Perfboard. The design follows a modular approach in which all three converters share a common AC input stage while maintaining electrical and functional isolation at the output level. The AC mains supply is first conditioned using protective and filtering components, including 2A fuses for overcurrent protection, NTC thermistors to limit inrush current, and EMI filters to suppress noise and electromagnetic interference. The filtered AC is then converted into high-voltage DC using bridge rectifiers, and bulk capacitors are employed to smooth the rectified output, creating a stable pulsating DC bus that feeds all three converter sections.

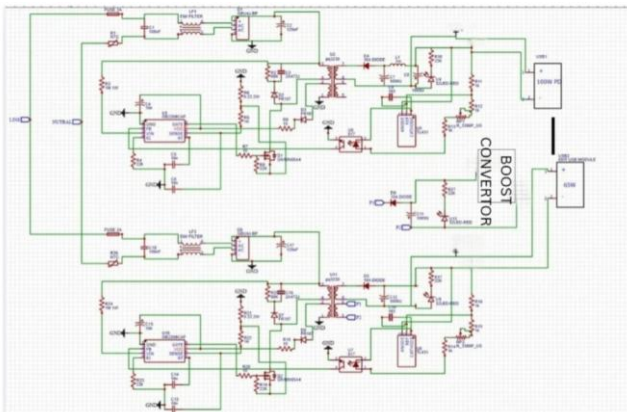


Fig.2. Circuit connections of 100W, 65W and standard charger on a single perfboard.

Each output channel is implemented using a flyback converter topology, chosen for its simplicity, isolation capability, and suitability for multi-output power supplies. In the 100W section, a PWM controller drives a power MOSFET at high frequency, which switches the primary winding of a transformer. Energy is stored in the transformer during the ON period and transferred to the secondary side during the OFF period. The secondary output is rectified using a fast recovery diode and filtered using LC components to produce a stable DC output of approximately 27V. Voltage regulation is achieved through an isolated feedback mechanism consisting of a TL431 precision reference and an optocoupler, which adjusts the PWM duty cycle to maintain a constant output under varying load conditions.

The 65W converter operates on the same principle but uses components rated for lower power levels. It includes a dedicated PWM controller, switching MOSFET, transformer, and rectification stage to generate an output of approximately 24V suitable for USB power delivery applications. Similar to the 100W section, it employs a TL431-based feedback loop with optocoupler isolation to ensure accurate voltage regulation and stability.

The standard output section provides a lower voltage, around 15V, using a simpler rectification and filtering the given

arrangement. This section is designed for low-power applications and may also serve as an auxiliary supply for control circuitry or peripheral devices.

All three converter modules operate simultaneously and independently, enabling the system to supply multiple devices without performance degradation. The use of separate transformers and feedback loops ensures that load variations in one output do not affect the others. Special attention is given to the physical layout on the perfboard to ensure safety and performance, including adequate spacing between high-voltage and low-voltage sections, short and thick conductive paths for high-current lines, and proper grounding practices to minimize noise and interference.

To enhance system reliability, several protection mechanisms are incorporated. These include input-side fuses for fault protection, NTC thermistors for surge current limiting, and snubber circuits to protect switching devices from voltage spikes. The feedback control loop inherently provides overvoltage protection by regulating the output within safe limits. Thermal considerations are also addressed through component spacing and appropriate power ratings to prevent overheating during continuous operation.

Finally, the regulated outputs are interfaced with USB-C power delivery modules for the 100W and 65W ports, enabling compatibility with modern electronic devices through voltage negotiation protocols. The standard output is provided through dedicated terminals for general-purpose use. Overall, the methodology demonstrates an efficient and compact design for integrating multiple isolated power converters on a single platform, achieving reliable, simultaneous multi-output power delivery suitable for modern charging applications.

### C. Solar charger design

The solar charger circuit is designed to efficiently convert energy from a solar source into a usable and regulated output voltage using a simple switching mechanism. The system operates on a low input voltage (around 3.3V), which is first fed into a control circuit built around a 555 timer IC. This IC is configured in an astable mode, meaning it continuously generates a square wave signal. The frequency of this signal is controlled by a resistor (R1) and capacitor (C1), which together determine how fast the switching occurs.

The output from the 555 timer (pin 3) is used to drive a MOSFET (Q1), which acts as a high-speed electronic switch. When the MOSFET turns ON, current flows through the inductor, storing energy in the form of a magnetic field. When the MOSFET turns OFF, the stored energy in the inductor is released and transferred through a diode to the output side. This process effectively boosts the voltage, making the circuit function as a **boost converter**, which is especially useful when the solar panel provides a low voltage.

The diode (labelled 4807) ensures that current flows only toward the output and prevents it from flowing back into the circuit. A capacitor at the output smooths the pulsating

voltage into a more stable DC output suitable for charging devices. The inductor plays a key role in energy storage and transfer, allowing the circuit to step up the voltage efficiently.

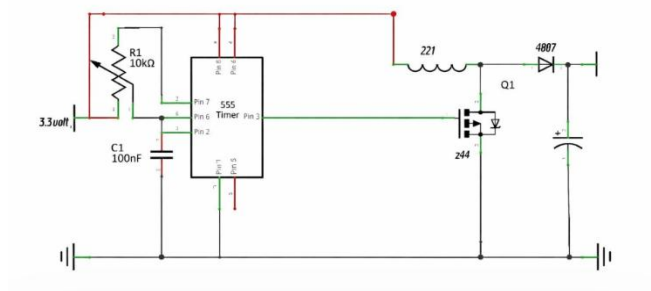


Fig.3.Circuit connection of solar charger on perfboard.

By continuously switching the MOSFET on and off at high frequency, the circuit maintains a steady output voltage even if the solar input fluctuates due to changing sunlight conditions. The simplicity of the design makes it compact and easy to implement, while still being effective for low-power solar charging applications such as portable electronics.

Overall, the methodology involves generating a switching signal using the 555 timer, controlling power flow through a MOSFET, storing and releasing energy via an inductor, and rectifying and filtering the output to produce a stable boosted voltage from a low-voltage solar input.

#### D.Comparison between chargers

The four charger systems—namely the 100W charger, 65W charger, standard wired charger, and the solar-based charger—differ significantly in terms of power capability, design complexity, energy source, and application. The 100W charger is the most powerful among all, designed to handle high-load devices such as laptops and workstations. It uses a flyback converter with precise feedback control, ensuring stable output (around 27V) even under heavy load conditions. Due to its high power rating, it requires more robust components, better thermal management, and careful layout design, making it comparatively more complex but highly efficient and reliable.

The 65W charger, while similar in design to the 100W version, operates at a slightly lower power level and outputs around 24V. It is suitable for medium-power devices like tablets and compact laptops. Because of its reduced power requirement, it uses relatively smaller components and generates less heat, making it easier to implement while still maintaining good efficiency and stability. Both the 100W and 65W chargers rely on AC mains input and use advanced feedback mechanisms (TL431 and optocoupler), which ensure accurate voltage regulation and safe operation.

The standard charger (around 15V output) is much simpler in comparison. It is intended for low-power applications and does not require complex control circuitry like the higher power chargers. Its design focuses more on basic rectification

and filtering, making it compact and easy to build. However, it lacks the advanced regulation and high efficiency seen in the 100W and 65W designs, which limits its performance for demanding applications.

In contrast, the solar charger operates on a completely different principle and energy source. Instead of relying on AC mains, it uses a solar panel as its input and incorporates a boost converter controlled by a 555 timer and MOSFET switching. This design allows it to step up low input voltage (e.g.3.3V) to a usable output level. While it is environmentally friendly and suitable for portable or remote applications, it is less stable compared to wired chargers because its performance depends on sunlight availability. Additionally, its power output is relatively low, making it suitable only for small devices.

#### IV.RESULTS

The experimental results of the proposed multi-output charging system demonstrate the practical performance of each charger module under real operating conditions. The practical implementation of the proposed system on a single perfboard provided important insights into real-world performance and design challenges. Unlike simulations or PCB-based designs, the perfboard setup introduces factors such as wiring resistance, parasitic capacitance, and layout limitations, which slightly affect the output performance of each charger module. The system was tested by connecting practical loads such as mobile devices, battery packs, and resistive loads to each output, both individually and simultaneously, to observe voltage stability, current delivery, and overall behavior.

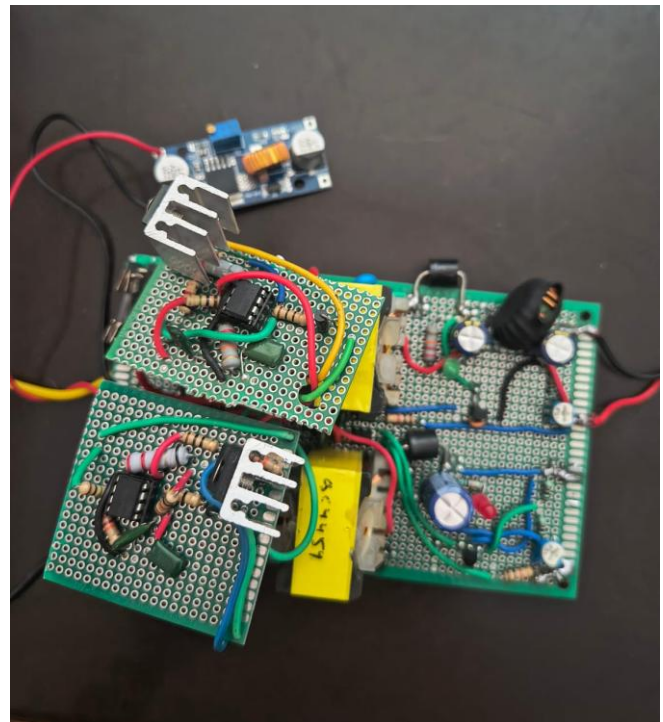


Fig.4.Experimental implementation of 100W,65W and standard charger on a single perfboard.



For the 100W charger, the measured output voltage was approximately 28V under load conditions. It was observed that the output remained relatively stable even when the load was varied, indicating effective operation of the feedback control loop. However, a slight deviation from the theoretical design value was noted, which can be attributed to switching losses, transformer inefficiencies, and voltage drops across rectifier diodes. Under continuous operation, the charger demonstrated consistent performance, although thermal buildup was observed in components such as the MOSFET and transformer, suggesting energy losses at high power levels.

The 65W charger delivered an output of around 22V, which is slightly lower than expected but still within a functional range for charging applications. During testing, it showed good load regulation and was able to charge devices without interruption. The reduced voltage level can be explained by conduction losses, component tolerances, and the non-ideal layout of the perfboard. Compared to the 100W section, this module exhibited lower thermal stress and more stable operation over extended periods.

The standard charger output was tested primarily for battery charging applications. It provided a stable low-voltage output and successfully charged connected batteries. Due to its lower power rating and simpler design, it showed minimal voltage fluctuation and negligible heating, making it reliable for continuous low-power usage.

The solar charger module was experimentally evaluated under varying light conditions. It was observed that the output voltage and charging capability depended directly on the intensity of sunlight. Under sufficient illumination, the boost converter circuit was able to step up the input voltage and charge a battery effectively. However, under low light conditions, the output dropped, highlighting the intermittent nature of solar energy. Despite this, the circuit proved to be a viable alternative power source for small-scale applications. This can be used to charge various devices which has various applications.



Fig.5.Resulting in charging the devices.

## V.CONCLUSION

This work presents the design and implementation of a compact, multi-output charging system that integrates 100W, 65W, and standard low-power chargers on a single platform, along with a solar-based charging solution. By combining multiple flyback converter modules with a shared input stage, the system is able to deliver stable and efficient power to different categories of devices simultaneously. The use of feedback mechanisms ensures reliable voltage regulation, while careful layout and protection features improve overall safety and performance.

The comparison of the four charger designs highlights the strengths of each approach. The 100W and 65W chargers provide high efficiency and are well suited for modern high- and mid-power devices, whereas the standard charger offers a simple and cost-effective solution for low-power applications. The solar charger, although limited in output, introduces a sustainable and portable alternative by utilizing renewable energy.

Overall, the proposed system successfully demonstrates how multiple charging requirements can be addressed within a single compact design while also incorporating an eco-friendly solution. This makes it suitable for a wide range of applications, including home use, travel, and remote environments where both performance and energy efficiency are important.

Another important outcome of this work is the validation of perfboard-based prototyping for complex power electronics systems. While not as optimized as a printed circuit board (PCB), the perfboard implementation shows that with careful design, proper component placement, and good wiring practices, it is possible to achieve functional and reliable results. However, the study also highlights the need for improved layout techniques, better thermal management, and reduced parasitic effects for higher efficiency and performance.

Due to the compact nature of the perfboard and limited heat dissipation paths, components such as MOSFETs, diodes, and transformers experienced noticeable temperature rise under load. This highlights the importance of adding heat sinks or improving airflow in future designs. The perfboard also limits the use of large copper areas for heat spreading, which is typically available in PCB designs.

In conclusion, the proposed system provides a flexible, cost-effective, and scalable solution for multi-device charging. It successfully bridges the gap between high-power charging requirements and eco-friendly energy solutions. Future improvements can focus on PCB-based design, efficiency optimization, heat dissipation, and smart control integration to further enhance system performance. Overall, this work demonstrates a practical approach toward modern, compact, and sustainable charging technologies suitable for a wide range of applications for various range of the devices and also helps in maintaining green energy.

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